

ActInfLab 2021 Quarterly Roundtable #4

Roundtable | 1:17:48

hmm

[Music]

[Music]

hello and welcome everyone to the academ

flab to our quarterly round table number

four that was our

first ever playing of a musical intro by

d schwakoff so thank you for that and

happy birthday to yvonne

so welcome to this

fourth lab meeting to our end of 2021

discussion not quite but almost

welcome to activelab everyone

we are a participatory online lab that

is communicating learning and practicing

applied active inference you can find us

at the links here on this slide

this is recorded and an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here

and we'll be following good video

etiquette for live streams so just make

sure to raise your hand in jitsi because

no one will be called on but if you want

to add anything please feel free just

raise your hand in jitsi and we will get

to it

today the way that it's going to work

we're going to address the lab scale

we'll have introductions then we'll talk

about some updates at the lab scale

and then we'll go into spending most of

the meeting discussing the updates at

the organizational unit scale and then
talk about the future but first
we can say hello and just introduce
ourselves

i'm daniel i'm a researcher in
california and i'll pass it first to
dean

hi my name is dean i'm in calgary canada
and i'll pass it to
david

yep i'm in the mountains of the
philippines and um

[Music]

pumped up that i can stop writing code
in a new language and write code in
another new language for a while
preferably sumo

yvonne

hello my name is iran i'm in moscow
it's uh

i'm a researcher in system management
school and an active conference club
um alex or stephen

hello i'm stephen i'm here in toronto
um

just in north york just north of toronto
at the moment

and i work a lot with um

embodied sense making and spatial
approaches

and

community development and

i'll pass it over to alex

thanks

yes i'm alex i'm in moscow russia and a
researcher

in active inference and system thinking
indulges so thanks

thank you and rj if you would like to
say hi yeah sure uh i'm rj i'm in new
york in the u.s
uh i'm a non-resident fellow at the
atlantic council and i do research on
sense making and knowledge management
well thanks everyone for joining this
will be a fun discussion maybe some more
will join in um and anybody who's
watching live can just ask a question in
the live chat
we're going to start with the
updates at the lab scale
so actin flab updates
we started this year
with
a
interactive
and participatory mission slash vision
slash project
which was to
ask whether active inference could
develop into a community and a field and
all that comes along with that
while also
drawing on the best practices
that we saw elsewhere especially in
systems engineering
and
community of practice that yvonne and
alex had a lot of familiarity with
and other experiences that we and every
participant brings to the table
uh having a community for active
inference that was participatory and
accessible for those with many different
backgrounds and trainings
having a body of knowledge that includes

courses and competencies things we'll
talk about soon
and also
reflected the nature of modern work and
professionalism and collaboration so
these are just a few of the things that
we wrote down long long ago
and it's still a living
document so please come get involved
all actions are participatory stephen
i was thinking we could also say a
community of communities a nested
set of uh
communities uh inside a community that
overlap in the kind of
markovian sense
yes
let's
graphically look at that
so this is a little updated graphic and
i hope it looks good for everyone
so on the right side in the gray
background we have the structure of the
active lab itself so this is the lab
scale that we're addressing in this
section
we have three
primary organizational units which are
coms communications edu education and
tools tools
we also have dot admin but they won't be
presenting on projects today
also in the active lab we have some of
the pieces of the puzzle that we have
gleaned from systems engineering
community and other areas
and we're going to be talking more about
the advisory board soon

and about other initiatives that we're undertaking so that's sort of inside the lab this is the lab scale under the lab blanket we have our organizational units dot coms

is

outputting different products into the broader ecosystem like externalities or runoff and so that's kind of this teal scroll image and coms is really our primary interface with the external ecosystem

we can think about this external ecosystem as containing all kinds of different things in it

so there's

artifacts there's texts

like papers and videos and all kinds of other

uh

documents and artifacts and then there's also people and their skills and their connections

so we can start in with this red bounded box of the current body of relevant publications which as we've seen stretches

back

and forward and left and right so there's a lot of relevant publications but there is a relevant body of publications

and we have used the live stream format kind of like a journal club to go into the publications themselves each live stream is centered around a publication which we'll talk more about in comms

comms produces a guest stream which bridges from that circle of established knowledge and people into broader interactions amongst participants in the topics that they're interested in like a two-way street between let's just say active inference and field x so some people are going to be hearing about active inference for the first time others are going to be inside of active inference hearing about x for the first time and then we look forward to having increasingly broad avenues to communicate like org stream that help to bridge even one step further and outside of the realm of just research and development may be thinking more about practice and all the different ways that people around the world are engaged in practices every day that may be related to or influenced by active inference so here's kind of this nested ecosystem of uh circles like an onion around the active inference literature sensu strictu and then all the ways that our outputs into the ecosystem help to scaffold and modify and catalyze sense making in that environment which can be overwhelming also which we uh picked up on from the systems engineering practices and community organization we've engaged in several development initiatives which we'll talk more about in the sections that are relevant but the ontology working group is uh upskilling and

learning about ontology and the systems
thinking reading discussion group is
doing similar for systems thinking so
we'll talk more about those but this is
just

our initial forays

into some of the kinds of products and
lab functions that are important and we
hope serve a positive role inside the
lab and effect a positive outcome
outside the lab in our local niche so
there's our internal states

over on the right side and then there's
our niche modification

products

in the left side

any comments that somebody would like to
add here

uh stephen

yeah i suppose

i like to think about how

the ontology working group is is

hopefully helping to bridge

and fill a gap in the in the complexity

of

um

all the different ways that active

inference gets used so it's kind of

hopefully going to help for that applied

active inference to be

more multi-disciplinary

interdisciplinary transdisciplinary

no small task but i think it's a useful

endeavor thank you agreed

and there's the ontology itself um

as a product in the niche

so multiple representations are very
important for distributed systems for

complex systems this is kind of a
snapshot visual layout but we can also
look at more of a narrative overview
so just to recap
during 2020
viatgan at all that's alex
and others including many on this
discussion now
collaborated on a
paper that was released as a pre-print
on september 9th 2020 called active
inference and behavior engineering for
teams so this is as close to a
prerequisite for understanding the
narrative and the context of the actin
flab as exists at this point
after
completing that paper
we sent out an initial call for
collaboration with an active inference
lab
and that was at the end of 2020.
during 2021 we engaged starting at the
beginning of january in our first year
of active lab activities many of which
we're going to be talking about today
that's kind of our recap moment
um and then we're going to also be
looking forward to 2022.
so this is like a
like a sandwich we're going to have the
two pieces of bread right now
and then that will be our appetizer and
then we're gonna spend most of it on the
meat itself so it's an odd sandwich but
please what do you expect
in 2022 we're going to do a few things
similarly and a few things differently

so on the
similar but different wavelength
we are going to be continuing to
implement our projects and plans within
each organizational unit so that's what
we're going to discuss today in the
organizational unit sections it's what
we talk about every week when we meet
in the organizational units so that kind
of
homeostatic or functional operation of
the lab is the substance of what we're
actually doing and so that is going to
continue
we're also
bringing a few new ideas and structures
to the table and practices
so one is the inaugural advisory board
cohort which we'll talk about on the
next slide
and then a little bit on the um
not having their own slide of their own
but important ideas that we look forward
and call for participation around
formalizing the lab governance there
have been so many important and
interesting
governance questions that have arisen
during the past year
as well as
everyone experiencing similar challenges
in their own located in online teams so
that's something we want to understand
better
and then also as we learn more and
embody and enact decentralized science
or desai
everything that comes along with this

from who's doing research what questions
are they asking how is it funded how is
it communicated how is it verified the
whole research stack is evolving
so
welcome to the 2020s and we want to
understand
how we as a lab can
take the best
aspects that are emerging from this
domain of desai while not getting lost
in the technicalities or committing
grievous ecological errors
anything else that anyone wants to
address these were just a few of the
kind of top level points that we all
worked on when we were preparing the
slides dean
yeah i think this is a fantastic summary
and i think for people who are kind of
new to this or maybe
are
still constantly sort of evolving their
basic understanding of this
um i think what we're what we're seeing
right here on this slide
is
the tension between a lot of constant
updating which is which is kind of what
the active inference tool
um enables and then the piece about okay
so these are the things that we've
established the kind of top-down
consistent constant stable things
that we think um have some shelf life
and can carry forward so i think by
showing this sort of chronologically
we're sharing with the audience the fact

that there is a tension there isn't one
constant that is going to take over and
there isn't a constant updating mean
we're always throwing out the baby with
the bathwater and i think
you may not necessarily
hear or see that in a timeline but i
think that's front and center in terms
of how we organize ourselves we organize
ourselves with the
with the intention of remaining in a in
a in a tension that constant updating
versus what's stable and is carried
forward
thanks dean and welcome blue great point
and it is like the nesting of models in
active inference where there's a slower
tick at the deeper
scale of the model so every year there's
12 ticks for months and then within a
month there's more and more
and that takes us to the advisory board
which is something structurally that
we're introducing in 2022
so
this lab advisory board first we can say
it's not a formal voting or governance
hashtag not legal advice not financial
advice they're not our managers
it's a advisory board
the advisory board is helping to move
the temporal scale of the labs
organizing from the kind of day-to-day
sometimes it felt like we were just
putting out little surprises all the
time
to the annual scale and beyond so as we
think about deepening the deepest level

of our model

for vision and for alignment while also

refining the finer points in our model

that's kind of like the iterating is

like the micro scale and then we have

deeper levels like semantics of vision

so that's kind of the way that we think

about these temporal scales

and we're already in the process of

onboarding individuals with expertise in

a variety of important domains

but please contact us if you have any

interest before

say the middle of december of 21

and you might want to be involved in

this inaugural cohort of the annual role

at this point for the advisory board

or we'll be there to

onboard you in 23.

what we do when we onboard them

in the advisory board just to be clear

and because many of them will be

listening to this

discussion

first we make personal contacts with

organizers and have consensus amongst

lab organizers about that invitation

and we'll have more governance for that

future

we then

have a live meeting with a potential

board member and assign them to one or

more organizational units so

no side doors no free floating

i'm here but where everybody is in an

organizational unit

and then we describe the specific

synchronous and asynchronous affordances

and roles and requirements that the advisory board members will engage in so it's not just an open-ended ask for you to do work it's actually a very structured conversation that will hopefully be win-win so we're really looking forward to the advisory board's service does anyone want to add any comments on the advisory board and we'll probably hear more from them once we've confirmed the personnel but right now it's just a um we're still in the onboarding phase okay any other lab scale comments before we go into the organizational units okay our first organizational unit as a educational nonprofit is education what we're going to do for each of the organizational units is we've just copied from our coda document which we'll talk more about we've just copied the ongoing projects so the idea is these kinds of single source of truth and centralized ways of working together allow us to have an asynchronous lab and then also when we need to present and share we just take screenshots or copy out tables so that keeps it really efficient these are the six projects or five projects that we have in edu so these are not the total scope of all the projects happening by participants these are within the organizational units

markup blanket

the goal of edu is to scaffold the

participatory and dynamic active

inference body of knowledge

so these five projects are the current

active threads and if you see a space to

improve one of these or add another one

then

get involved

so on to the first

one of these projects this is the

fristen symposium transcript so i'll

summarize it and then dave if you'd like

to add a comment or anyone else

so in june 2021 we had our first applied

active inference symposium with

professor carl fristen

and each of the three organizational

units had about a one hour session

the videos are watchable

and many insightful things were said and

discussed

so the idea of this project is to use

automated speech-to-text processing plus

manual copy editing and enrichment of

those texts to get an accurate

transcript of the discussions so that we

can upload it as a preprint get a doi

and make it citable so instead of oh i

think somewhere somebody said this thing

let's have in-text something we can copy

out

and connect core terms to translations

use it for the captioning of the youtube

videos for accessibility

and then looking beyond this one

important symposium we had this year

towards transcripts for maybe all live

streams
and integration more
thoroughly with the ontology so
translation of terms linking to other
knowledge resources so that's the
fristen symposium transcript dave or
anyone else do you want to add a comment
yeah um

the
three
transcripts are um in daniels and blues
and some other people's hands
um
and um

[Music]

as much markup as folks want to do
that's fine if you want to just take
what i've got and make them perfect and
get all the cross references and so on
that's great if you want to just um
mark up uh
the change of speakers and maybe throw
in some obvious paragraph
terminations just empty lines that's
that's fine
um and i have the option later on of
just doing some some automated cleanup
uh trying to find end of sentences that
aren't marked otherwise
but um
you know i think this is going to be
somewhat quicker than just taking the
raw youtube
output well we shall see
uh next year after some other things i'm
going to go in and improve the program a
lot more do the stuff that i thought i
would have finished by now but

you know when i can't change the parser
and i have to live with the parts that
i've gotten it's only got half of the
vocabulary and i gotta get around that
it takes longer so that's good
if you can't pass the one you want
parse the one you have steven
yeah i was thinking that one thing to
think about when we have these talks
with um
kristen and these presentations is
we're able to as well as clarify the
stuff which is um
already possible to do
and separate it from
what is um
you know is is beyond our knowledge at
the moment we get this zone of proximal
development
the vaikovsky which our russian friends
will be
very fond of um so that idea of where we
can be we get a sense of where the sort
of the extension of what the established
knowledge might be and i think that's
also a useful thing
to clarify um because that's also part
of where this applications are going to
be
good point we're not just
recapping memorized facts
that's the first in symposium transcript
the second project in edu is the
ontology working group so in ontology
working group we are reading this
textbook by adam peace the ontology
uh a practical guide textbook
and the bigger picture here is that

there's a continuum
of formality and expressivity of what is
called ontology ranging on the left side
from mere terms lists
through more structured and enriched
terms lists like structured glossaries
and informed hierarchies
verging towards
more formal data structures that allow
increasing computational
power
in many different dimensions
and over here
the sun rising in the east of sumo
and uh sumo's the language and the
framework that's discussed in the
textbook suggested upper merged ontology
so that's what we're learning about
in the ontology working group and we
meet every two weeks
we discuss what we've read and worked
through the exercises together
and
over the course of the year the ontology
working group has been kind of pushing
like a little train
towards further and further right
from
um
just the terms list that we started at
in the beginning of the year
to a lot more which we'll talk about
soon but the ontology working group is
specifically those who are learning
ontology by doing and reading the
textbook
any
thoughts or comments on ontology working

group and we'll talk about the ontology
more broadly in just a few slides
okay
so another
development initiative
which um i'll allow yvonne to summarize
is the systems thinking reading and
discussion group so yvonne maybe you
could summarize the intention or the
background here
yes thank you
well
in initially the system thinking course
was represented by system management
school in moscow and we decided to
provide
a common language for the lab
all of us from
different domains with different
perspectives
and we decided that it will be
somewhere we can
understand
one another better
here i
gave some small example when
where one
object can describe
a totally different
one person can hear can see here a face
and another
vest
and once
all of them
talk about leaps they totally
to
miss
understanding

and to prevent this
um situation
this group
could you could you
texts
next yes yes
there is the interface
of the course
is this is uh online course based on
iso standards and for documents
that
include system approach
in engineering and
uh
in different
a lot of
fields
and
of course
the course
has a
list of concepts that
you can actually interconnect it with
the active inference
concepts
and
every every participant
can now
uh upload it to the course
and
in
appropriate
way
start to learning system and
thinking as
system management school providers
of course has
eight sections now a lot of tasks

to
check
your understanding
you can find all information
about the course on systemschool.com
thank you yvonne and also you and others
translated it from the russian to this
english course which were beta testing
so it's being offered for free and we
provide them feedback so it's a cool
relationship between our groups that
helps benefit our participants
any comments on st rdg
one of the main projects in edu
something we spend half or more of each
meeting on is the actual active
inference ontology itself
so here we're showing um just the top
page of the working active inference
ontology we released versions um more
stable versions
earlier in the year
um
we're working through the core terms
and then for each core term getting a
bunch of references and quotable citable
definitions so not saying that they're
the truth just how it was used you know
a system is a or blanket states are
then
each week we have a discussion
on these terms
one term per discussion and we kind of
go deep and hear many different
perspectives and gather more resources
and then we work towards
proposed
synthetic definitions that synthesize in

the term of synthetic not artificial
that bring together a few of the senses
or clarify is there just one sense or
are there multiple senses that the term
is being used
and then the terms themselves
as well as how the terms are connected
to one another
will form some of the early versions of
our educational materials
so how about that five minute video that
everyone has always wanted about what is
a markov blanket or
that five minute video about how is a
generative model
related to epistemic and pragmatic value
those are the kinds of questions that we
want to have like a video
but more importantly this deep knowledge
linking underlying so just that's the
first cover sheet and then i'll get to
you steven um the second sheet
has translations of those core terms
into
human languages
so we have russian portuguese spanish
french and italian
very well completed
we're definitely looking for more
languages to be added so if you're
speaking or learning
any language
then it would be awesome even just to do
two
or five words terms because that helps
modify the niche so that stigmatically
another language speaker may be able to
pick up where you left off so this would

be extremely helpful and important
global work
for somebody who speaks one of these
languages or any language to start to
pin together
just with the terms at first this active
inference corpus which is primarily
written in english as a natural language
mathematics and computer languages as
technical languages and so we'll be able
to
operate better across those different
human and technical languages and then
the last sheets of this tab
here just show the supplemental terms
and equations so the equations that's
going to be a long game with
corralling the formalism
and how that has evolved and how
different letters have meant different
things through different times but then
also supplemental terms there's so many
fun terms that come up in our
discussions but they're not necessarily
core or they may be we may
update them and put them in the core
later but
things that people bring up
that like
are cool and important to learn about
and link around we can add that to the
the term supplement so we kind of get to
have a big gutter with a term supplement
while also having a very
curated core terms list that will be the
basis of education so stephen
yeah i'd just like to also thank again
that ivan and alex there for bringing in

that system's work

i think it's really valuable and i think

this

as you mentioned there the different

languages are also useful because

even different languages have different

ways of thinking about time they have

different cultural so if we're looking

at pluralism

um but also trying to clarify and get

ways to have

common languages this is going to be

actually extends that

and one other thing i'd like to mention

is that

it can build the bridge with the field

of coaching i think that coaching

systemic coaching and ontological

coaching are two very fast growing

fields and i know from the literature

that they're really moving into looking

at um

pluralism

and different types of social

construction

and i think

so this this

before it's a bit of a bridge too far

but i think with this work as well from

alex and ivan

and this other work with the ontology

that

is starting to make it not bridge too

far hopefully it's it's it's spannable

thanks

very cool

the last project to discuss in edu

is course development itself you know

what people expect of a edu to engage in
and we know this is one of the most
important projects and we also knew
that ontology
not necessarily the practice of ontology
itself but this core term ontology would
be the backbone of the course
development
broadly in this project were interested
in how to provide the affordance for
people to learn and apply and
communicate active inference
we had great discussions during the year
on live streams and in edu
about instructionism and interactionism
and how do we prevent this from just
being
a sort of plug and chug course how do we
enable interactions and
his own approximate development like was
being described and this tension that
dean described earlier as well between
the sort of
the technical the minute particulars and
the big picture and the unknown and the
thirdness and the betweenness
and um with a special focus on
applied active inference and also
enabling this education for individuals
of many different backgrounds
inside academia outside academia there's
a whole world out there and there are
not paths from here to there so that's
what course development is going to be
exploring
in 2022 and beyond especially
everyone is welcome to get involved with
all projects but in edu if knowledge

engineering sounds interesting to you
whether it's the translations or diving
into the literature to find definitions
and references
bringing your experience on curriculum
development
or learning by doing on the ontology
those capture a few of the projects that
we talked about here and that's some of
the main practices that are involved in
edu
any dot edu thoughts
okay
dean
oh wait unmuted then continue
dean
sorry about that there's some cultures
that
we're taking a very western-centric
approach to this which is fine we have
to start somewhere and i think that
there are some cultures that simply
won't have words
or verbs that describe some of the
things that happen in their
particular niche that those with a
western view don't necessarily have the
terminology for and i think we're
keeping that aspect of this open when
we're trying to figure out whether or
not there can be a bridge from calgary
to moscow or if there are other ways
of being able to transport ourselves and
migrate that are better than trying to
put a solid structure in
so sometimes when we're working on this
ontology piece
we might only be able to collapse to two

or three different ways of describing
two faces and a vase and i think that's
one of the other things we we want to
let people understand is that there is
no one
singular answer to this that when active
inference is actually being used we
distribute and there are going to be
times when
we can agree to disagree on how to
describe something
thank you dean it's like there's
what was said in first in 2013 the paper
there's how you just said it right now
there's how blue's gonna say it in six
months
and those are like observations those
are not the latent hidden state even if
they all concur that isn't alone
evidence that that is the hidden state
so we actually can take that kind of
signal processing
perspective and integrate it with ideas
like pluralism and distributed systems
to talk about how to have inclusive
education
so in comms the second
organizational unit we are interested in
communication
the goals of this
organizational unit are to organize the
internal lab project and activities kind
of the connective tissue circulatory
system
and also to carry out all forms of
communication with external entities so
if.edu needs to send an email they can
just pass it to coms and say please

email these nine people with this
information
thank you
and dot coms loves emails
the first project well there's uh six
projects on here
live streams social media podcast end of
2021 treat monthly newsletter and
updating the coda and the playbook so
first on the live streams of which we're
in one now
we had
a fun year of live streams the live
streams technically began before octave
lab in july 2020
but
we have um
multiple series
of live streams and we've done
well over 100
total 85 around papers and then 14 guest
streams presentations
seven
model streams and three math streams
which are more technical
and then um
roundtable meetings like this one
orgstream is a new series that just
began we have one video in there about
organizational design practice and then
the applied active inference symposium
and
lest anyone think the year is over
we still have some upcoming papers to
read
for 32 still cast the chaos markov
blankets in the first two weeks of
november 33 thinking like a state with

avel serval

33 in the last two weeks of november and
then we'll have two more papers probably
that we'll discuss in december but just
haven't set them yet

so we'll also be probably moving this to
coda having a nice
way to look at the past and present live
streams

but for now

if you want to give a live stream

like give a presentation and or
discussion for a guest stream on your
research or your thinking or your
application

just let us know

and also if you want to suggest somebody
who would be good for a live stream
whether it's just like i read this paper
could you reach out to this person or
whether it's somebody you know and you'd
like to co-organize the event or just
make the connect we're planning for 2022
so that would be great if anyone
listening has any thoughts or wants to
present

social media

um

[Music]

broadly just to give two updates we have
over a thousand twitter
followers and uh over 600 subscribers on
youtube

so

the slow and steady

accumulation

there

on the podcast front uh then i'll pass

this to blue so anything you want to add on social media blue and then please describe the podcast

sure so just also on the social media front um i just kind of took over uh the facebook page and um or or took responsibility for you know promoting the actin flab on facebook so if you're on facebook um find us there because that's brand new and we have like i don't know a whole 10 likes on our page or something like that but but we've hooked up the podcast which will feed directly into the facebook page so if you get updates that way that's helpful and we'll try to post the live streams on there as well

so

yeah new social media venue is always good we're also on discord right so um if you're on discord and and do that too um there's more social media ways to get a hold of us

and then podcast i think we're up to like 30 episodes of the podcast at this point and we're starting to codify them um with links to each episode and the associated live streams and maybe in some future rendition of the website we'll be able to make all that information publicly available so that people can see who participated and what the topics are because frequently the podcast will just be like the markov blanket um even though the paper is is a much broader

paper or topic so the podcasts are coming along and yeah get in touch i

mean there's always room for people to participate i've been doing a lot of them um also we're looking for music so if you want to submit some musical riffs to us uh for evaluation um definitely that's something on my on my to-do list is to like get an intro lead for for the lab podcast and if if you're interested in podcasting or audio production at all um get in touch it's been i've just kind of learned and figured it out but i'm happy to you know work with people show people the the ins and outs of that so yeah awesome job with these edits blue because sometimes the discussion can have even when it has just a few core themes which sometimes happens it can be interleaved complexly so it takes skill to untangle but the podcast is fun um

[Music]

number four the end of 2021 treat so this is just a little gift that blue and i wanted to send out so this is kind of our project we just thought um we could send people some physical stickers and potentially send a non-fungible token an nft purely as a gift no value or utility so both of these reflect the expansion of the lab into the physical space as well as the digital and the cryptographic space and we do think that cryptographic affordances will be very important for

the future of science art and
communities in web 3 and beyond
we also want it to be
mindful ecologically and meaningful for
participants so
if you'd like to be involved in this
area
of kind of
taos and web3 and crypto and desai
then
please come get involved with active lab
because there's a lot to
learn and help us do here
just very briefly we have a monthly
newsletter so that's um
you can find that on our site
activeinference.org
and find other places to subscribe but
this is a kind of monthly and then a few
other special
affordances but mainly just a monthly
summary
of our
operations so good low traffic totally
not spam way to stay updated
and then the last comms project
is um the live stream playbook so we do
have a slides-based playbook
for
uh live streaming so i'll just kind of
like show it here
this is just
for participation guidelines
this is um checklists for roles
participant checklist presenter
checklist broadcaster checklist
facilitator checklist organizer
checklist

so that was something that we developed
over this year
and we're looking to
move that over to
coda
so that it can be implemented much more
smoothly with automations
the playbook itself from the technical
details to the behavioral and the social
and then also another coda related
improvement project is going from the
list that spreadsheet of live streams
to this enriched table
where we're gonna be able to add in
keywords links participants
um you see some upcoming streams here so
it will be possible to very easily ask a
question like i want to know what dave
had to say about predictive processing
before 2022
that will be a simple query to make
and when we include the speech to text
and have full text transcripts not just
enriched tagging that'll also be very
powerful but this will be really a fun
project and it's also little micro tasks
if anybody wants to contribute
asynchronously like i loved number 14.2
i'm listening to it already i'm going to
write down a few keywords that were
spoken that's a huge way to help the lab
think about how many times a podcast or
live stream is listened to
and no one enriches it or makes it
searchable so this will help improve
that situation
and
connect these resources to each other

so if you want to get involved in dot
coms if you like communication and you
are familiar with active inference or
not of course all activities are open to
everyone

special roles

or practices that could help a lot would
be somebody who wants to like lead and
grow and automate active lab social
media accounts because that's not really
our

jam

also if you want to get involved in
co-organizing or facilitating or

broadcasting live streams

introducing new series compiling and
curating research products onboarding
participants

new kinds of communications like blogs
short writings tutorials there's so many
things that we can

do in comms and if you just think that's
a communication i

expected to have existed i i expect that
five minute video on markov blankets to
have existed or i would prefer it
existed

we'll connect you with the affordance so
that you can reduce your free energy by
doing it and also helping people any
comments on comms

okay final organizational unit dot tools

so in tools the overall goal is to

enable the effective tool and instrument
use for all active lab processes

the second goal is to explore and design
affordances for our niche resulting in
effective action as well as innovations

and tool development so it's a lot like
comms there's this interior
connective tissue role and then there's
an external side in tools the internal
side is our tools what we use in the lab
and that's what the participants are
engaged in because we're all
participants here
and then there's like
this medium and deeper question
of
what do active inference tools look like
recommendation engines and robotics and
all these kinds of things and we have
explored that in some depth during the
year but let's go over the projects
themselves
so first is the cloud infrastructure so
this is an internal but eventually we'll
have an external component as well
we're looking to develop accessible and
powerful computational resources for the
lab
there are several use cases that are
very important
so
some of them are admin tasks like having
the website emails file storage crm like
personnel management
in um a secure cloud environment
there's also use cases like a
personalized learning dashboard or
workspace so somebody can just log into
their active lab accounts and have their
resources and learnings right there for
them in a high attention and high trust
setting
two special use cases that are

can be mentioned here

one is ontology development and natural language processing so sumo it's an ontology but it's actually written like a text program kind of like lisp so having an environment with sigma and a few other pieces will allow people who might not be able to set up these often esoteric programs on their own computer

will have the cloud image virtualized and ready so that you can play around in this sandbox where all the toys work

and then similarly on the machine learning side although much of early work in this vein was done in matlab in the spm statistical parametric mapping toolbox increasingly modern development is happening using the languages of python and julia and we spoke with people in live streams and guest streams who talked a lot about that and so we want to have a

what's a fairly common setup in industry and in computer science research labs to have a cloud interface with for example interactable notebooks that helps you do machine learning in the cloud without having hardware or software installed locally so that's kind of the cloud infrastructure side and of course these aren't done projects these are things that people are involved in and can be involved in feel free to just raise hands if you

have any other thoughts but um another
project is the actin agent development
so we
here are curious with
what does the future of active software
agents look like can we make an
interoperable
software
package for active inference agent
modeling
and where could that be applied we
started with the nethack challenge which
we didn't muster the regime of attention
for this year but it was ambitious and
we had many many
good discussions and we made a lot of
useful
advances we just
didn't end up doing this very
challenging
machine learning competition
but
there'll be so many cool areas to apply
active inference to
so that's kind of like just like people
expect the edu to make courses but it
does a ton of other stuff too
tools sounds like it should be making
something like this and we'll do other
stuff too
project three in tools is the
participant onboarding um yvonne do you
wanna add any detail here
yeah as then already mentioned we
this quarter
extremely
more use
code

tools and here we
as
other
other our activities we
model
participants
as
a person who can
simply
be involved in different projects within
the lab
and
as
other
our activities we based on systems here
and
for
different scale of the lab of project
and the tasks
within the project we have
is not not a simple but
we have a model that
we use to
to simplify the onboarding
so within the project we overall and
we
opened to
assign this role to performer
with the skill we understand we need to
make this role
and
each role
has been in organizational unit of the
lab and to
perform the task
we provide some
um
[Music]

some information to

[Music]

the information to to have

understandable understandable task

with

any any skill you you have already

thank you yvonne

and

that relates to the crm integration

which is

usually using to refer to

customer relationship manager but

they're kind of like personal

relationships

and

it's a project that is on coda

it's under the dot coms but the

integration is a tool's concern and so

just like yvonne laid out here

using these ontologies for projects much

of which are derived from systems

engineering and systems thinking we'll

be able to have everybody entered into

the crm and then it'll say okay

blue and i are going to be doing this

live stream

they added us to a calendar the template

is here the papers here the emails in

the inbox

that will allow um very flexible

coordination

and also improve the accessibility

because not everybody is going to be

trying to um

prune every part of the garden

each person will be able to have a

targeted regime of attention that

is according to their preferences

and also will be contributing to
something bigger so these are projects
involving
making the organizational unit
engagement more accessible
project five
uh yvonne again please describe what is
the uh systems thinking reading and
discussion group which was an edu
project on the learning capacity what
does it have to do with application in
the dot tools
so
as each project where
we can we have to divide the project
itself and
the project that enable this project so
to provide
all participants with
all needed instruments we need to
enable it
and uh
for
more understanding of
the course and the discipline
of system thinking
uh we decided to
set up
a
reading and discuss
group
and
[Music]
one per two weeks
we plan to
meet and
discuss
what

what we
learn and what
what
hard
things we met in the course
uh and here i provided two
two screenshots
it's
that the first first of it is how it
looks like in coda
and
and the second one is the
backlog of the course
if
body
during the
learning
report issue on
what typos
typos it
the course has or
uh any comments they would like to
provide
and if you
move to the next slide
please
yes so
once once a student
record an issue we have
this table
and each issue
automatically
[Music]
placed to
this this table
and
um
because of

different

translators work

worked on the course

different

chapters

the

different translators

worked on different chapters

and each

issue

is assigned to a special translator and

within

the systems

system school we

we fix

and what what do we

need to fix in the course

thank you and this will be one of the

kind of templates and inspirations for

the active course that will be using

tools that are state of the art

which were selected from a rigorous

process for

the systems thinking course

a sixth project for those who

watch all the way to the end of lab

meetings you know who you are

is the graphical interface and

extensible knowledge graph project so

this is going to be something pretty

cool heading into 2022 and beyond on the

left side

on the left side is an image

that i'll just expand so it's a small

this is

within

matlab which is a proprietary program

that you have to pay to have a license

for
many academics are given access but
that's not everybody
within matlab the program
there's a package called spm statistical
parametric mapping which has been
developed by professor carl firsten and
many others for the previous decades
inside of spm
you'll find a function called dem demo
and it will bring up this graphical
interface
which has kind of sundry links to
some simulations some code examples some
visualizations some papers but much of
it is within matlab itself
so this is kind of
a resource that's been developed and
locally optimized within this
intra spm intra matlab
format
however one of our major projects for
the lab and the community moving forward
is going to be reimagining
and deploying
a graphical interface and an extensible
knowledge graph for active inference
so what does it look like to make a
graphical interface an extensible
knowledge graph that is participatory
with open source development and modern
affordances for knowledge graphs
that uses the active ontology as a basis
of the knowledge graph so for example
here it's sorted by keywords but you
know what do you do when one button fits
under two keywords or two keywords apply
to one button that's why this is a local

optimization because to have that kind of advanced and even personalized re-rendering it's required to have a more advanced ontological system than just terms or principled hierarchies which is exactly why ontology working group has been involved in this work and these practices

because the extensible knowledge graph starts coming into play on the far right of that spectrum

we also wonder if it can be deployed in a website

so not requiring you to have matlab install spm find dem and then maybe it works or maybe it doesn't

how about a website where there's maybe even computational resources so people can run simulations and save their own versions of the code in their own workspace

and of course

what else this is a project that we're barely just beginning

so if you're interested in

user interface or education or tool development something that would be an ontologically

scaffolded

knowledge graph containing very heterogeneous resources like

terms and educational courses papers videos etc

this could be

a very important

and also very active project so if

that's kind of sounding cool to you then get involved

overall

uh dot tools

has all kinds of uh well we'll just

delete this

dot tools has ways for you to get

involved if you'd like to contribute in

any way

let's go to the future and then of

course everybody please feel free to

raise your hand so the future

section is

short because we can actually just break

it down

systemically and systematically to use

some vocabulary that i learned from

systems thinking

into these different levels of

organization of our lab

there are the next step steps for the

lab

as an entity which involves for example

our nonprofit registration continue

continuity and the onboarding of the

inaugural cohort of the advisory board

there's the next steps for the units

inside of each lab which were discussed

for most of the meeting today

there's the next steps for the projects

and the teams that are inside of each

unit which have specific coda spaces and

timelines and roles and requirements

of themselves

and then there's

the next step for you as a participant

so

if you don't know that's perfect because

that means that you'll have a lot to

learn by doing and reduce your

uncertainty through action

so if you're listening to this or if
you're aware of it you're already a
participant you're engaging in the
regime

of attention regimes of expectations
related to acting of lab and
active inference more broadly

so

if you're here you're a participant

if you know what your next steps are
what policies you're going to select in
the near future or now that's awesome

if you do not know what your next action

states should be what policies you
should select or maybe even what
policies you can select what your
affordances are then

you can get involved by just

communicating with us so anyone can feel

free to contact welcome scott to the
very tail end

um we're literally just

closing up and uh just have about to
have an open

discussion where anyone can just
add any points

what was something that they
liked about 2021 what's something they
want to change what are they looking
forward to in 2022

anyone's welcome here

well i'm too i'm late to the game so i'm

i don't know what was said before but i

um i feel like the we're on the cusp um
of folks

um

benefiting from this kind of framing in

a much more broadly
and i think that was interesting i just
saw on twitter there was an interesting
article on the
rethinking of pristine um pearl blankets
versus fristen blankets discussion which
i thought was interesting and it's not i
haven't read the article yet but it's
interesting that notion of
the maturation
of the concept and whether or not that
critique is accurate
the critiques
are helpful because they start to bound
the
um framing and make it more useful for
more people because it's not as
over enthusiastic in its ambiguity on
the at the edges
and so i'm really excited about the
maturation of
both this concept and the communities
that cluster around it and i feel like
this particular initiative is going to
be really pivotal
in that and it's um i won't use the word
priesthood and priestess hood here but
um there's a lot of um
evangelization that will be
done in the in the next uh or
evangelizing i guess that will be done
in the next year
and i'm really looking forward to it i
think a lot of people are going to
find
that when they enter the space that
people are are for some reason seem
fairly reasonable given a lot of other

spaces and willing to
entertain the limitations and
constraints what we have here because i
think it is something that has
a core
of something that's new and offers
helpful insights so i'm really excited
about the coming year thanks
thank you
dean

yeah i think my takeaway from the months
that i've participated so far is that
one of the big differences kind of a
dramatic difference

if you will

is that i find that working in this
medium it's more like a cnc mill than a
than a printer a 3d printer because
instead of

a bunch of building up
we're essentially removing a lot of
noise

and trying to get to the idea of
what are some of the signals that we
otherwise wouldn't be able to see
because we have a chance to interact the
way that we

have and do i don't think it's uh
i don't know that it's necessarily a
consensus building

platform but i do know that it is
certainly a way of being able to
uh

cross pollinate

we don't tend to collapse to one
particular type here and
that that last

input being a perfect example of what

we're trying i think what we're
achieving here
and i think that the going forward piece
will be
really interesting because it isn't
always describable in this moment
i think a year from now when when we're
doing our fourth quarter a year from now
god willing
um i think
again the updates will
tell us a whole bunch about what we
learned in the meantime instead of um
trying to come to a particular
specific outcome and i think that's
probably one of the
greatest joys of participating in this
thanks dean
fellow chitzer with a raised hand
all that sorry
somebody who's muted but i'm not sure
who exactly it is
is it me you might yeah
okay um
yeah
i i following on from what was said
there is the idea of um
the blanket early on was kind of like a
part of
active inference but the having been
talking a lot more and you know thinking
about the fristonian blanket as
a particular way of thinking about
blankets and knowing that these blankets
are used in other
contexts i think has helped mature
things i mean scott's been talking about
the idea of maybe synthetic blankets in

other contexts and i think
what's quite interesting is i think the
discussion
um
around the nature of blankets is going
to be um
as a tool or maybe as part of the
ontology is going to
help at least
help separate out the discourse
um
and i think that's not a bad thing
because uh there's there's there's um
there's there's uh
there's markov chains and there's lots
of these processes are used in other
fields which are allied but are not
active inference um and then well what
happens to these when we come into that
field and uh i think it also then starts
to help with some of the pluralistic
ontological questions
allowing other parts to maybe be more
formalized and then the ontologies might
happen more
through the way that we think about the
realization of the blanket form and the
sensory states and the observation
states been interacting with
thank you stephen
so anyone else can uh
raise their hand definitely the core
terms yup scott go for it
so you know it's interesting steven just
got me thinking so one of the things
we're doing this nsf thing some of you
know about it right now and doing it
with some other folks in involved in

complexity

and um we're doing virtual uh excuse me
uh verified information environments and
you got me thinking stephen you know
credit card system

we say is a verified information
environment we're defining it as a thing
that has business operating legal
technical and social variables that give
it integrity

so it's interesting when you think about
blankets

in a way a verified information
environment what we're saying is there's
an externality

or an ex there's a an external
environment that can be named
and characterized so it's like before
there were countries nation-states
we didn't say they're nation-states we
had the feudal kingdoms or whatever and
it was a characterization of a set of
relationships

and it became paradigmatic only through
broad adoption so we say now there's a
society or we can say there's a museum
or there's a school each one has a
concept but it's really a set of
relationships not a building
and it feels like i wonder if markov
blankets it just started feeling like
perhaps

there's a naming of a set of
interactions a blob of interactions out
there that can start to build up its own
um not not rigor statistically but
cultural
um recognition

and and if and that i wonder
whether markov blankets
are candidates for that i don't know if
it's powerful enough a concept to me it
feels like it may be but and and that
set of practices that occur associated
with markov blankets
um i think can be elevated into best
practices standards and institutions
so we might actually see is the in
markov blankets for instance might
become institutionalized into fiduciary
layers
for representing parties things like
that anyway you got me thinking stephen
on that one
nice
thank you um
i was just going to say that the core
terms each of them
are so
rich and uh the opportunity to go into
technical details and discussions as
well as apply discussions
um
there's so much more that could be
said
the only thing i'll say is personally i
don't think that having names associated
with terms is helpful because it's not
referring to a person themselves and so
i think once we move beyond some of the
name based nomenclature will open up
into a more accessible and rigorous
space
not based around whether darwin really
thought this way or not about evolution
or whether markov which markov really

thought this way or not about statistics
we can just frame positively
indefinitely and probably very austere
what we actually mean
and that will enable translation
automation and many new things that just
last names don't give but that's going
to be part of the fun
i never even thought of the mark of part
as and everything occurred to me it was
a
guy's name but of course it is but so i
i vote for snuggly we should call it a
snuggly as a generic
term i think let's change all that into
russian what's a russian what's russian
for a schnoogley blanket what about
snuggie like the blanket with arms
it's already trademarked
let's change that frequency to snugly
um well um and then also one just mark
off no just
uh
we are actually unsure which markov is
being referenced this was some
scholarship by dave to actually address
the question which a markov the the
elder or the younger
are some of these terms referencing in
the english-speaking literature and that
is not always clear so
there are just
opportunities for international
scholarship that we discover when we're
curious
stephen
of course you've got the ian we can add
on so the fristonian i mean it is

problematic being his name but or the
markovian

um i mean that sort of opens it up to
the hand waving but it um
that may also sort of show that there's
more than one version because you know
you just hear markov blank it does
to someone who came for the first time
they may think well that's that's the
thing it's not it's that's it right
whereas

um once you get into markovian monism
um

and even that maybe that there's i can
see why carl wouldn't want to
necessarily call them fristen blankets
because then it gets into that whole
thing about it being his name
but there's there is something possibly
between
or beyond a markovian blanket that would
help denote what is going on with an
active influence blanket where it's
bi-directional um but this is this is
great conversation we'll talk more about
this obviously

yep

perfect um

[Music]

that's all that uh we need to i think
note for the lobbying but yeah thanks
for sharing about that these are the
literal discussions in the terms
discussions in edu every week these are
like the fun things that we get to
talk about so
um to those
in this jitsi many of whom were

um just totally
essential and non-fungible in their
contributions to the lab
as well as those who are
watching the replay and who participated
in every activity this year it was just
an awesome
year

who could have thought that one year ago
things would be where they are now
and

the game is just
beginning so we hope that
everyone listening whatever their
background or familiarity
with active inference or other topics is
they're just welcome to get involved
any other last thoughts otherwise
that's it for the fourth quarterly
roundtable meeting

okay
should i play the uh theme song again
okay you

i don't think you're gonna hear it
on the uh you're not gonna hear it on
the stream but so just
it'll be quiet for 30 seconds

[Music]

oh wait i need to restart it without the
little filter to do the voice okay so
here we go theme song is going to start
now and then i'll end the stream

[Music]

okay
thanks and bye